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Public course materials: <https://learn.cs.illinois.edu> · Course: <https://cs124.org> · Prior research group:

<https://bluegroup.systems>

Profile

I am an internationally recognized computer science educator and educational-technology researcher whose work has changed how introductory computing is taught at scale, combined with an earlier, internationally cited record in mobile and sensor-network systems research. I am now leading the redesign of computing education for the AI era—rebuilding a 2,000-student course around AI coding agents, creating new courses and assessment methods, and designing a new undergraduate degree. I build the software infrastructure behind my teaching, publish it as open resources used by learners and instructors worldwide, and have won my university's and college's highest undergraduate-teaching honors. Over my career I have taught more than 16,000 students, supervised 47 graduate and undergraduate researchers, won almost \$3M in competitive external research funding, and accumulated more than 8,000 citations to my published work. From May 2027 I am joining the **University of Sydney as an Associate Professor (Level D)**, bringing this work to Australia. My field—computing education and the tools that support it—is practiced and recognized internationally, and my contributions to it are openly available and demonstrably adopted beyond my own institution and country.

Computing Education for the AI Era (*primary field*)

I am redesigning computing education for a world in which AI coding agents write most of the code—across a 2,000-student introductory course, new courses for non-technical students, and a new undergraduate degree. This is novel, internationally visible work, and it has been recognized beyond my institution.

Rebuilt CS 124 around AI coding agents. Starting in 2026 I redesigned the larger project in [CS 124](#) so that **every student designs and builds their own unique Android app working with an AI coding agent (Claude Code) throughout**. When AI handles the translation from specification to code, the valuable human skill shifts from translating a specification to *formulating and directing* one. A scaffolded pipeline—paired brainstorming, rotating pitch feedback, AI planning sessions, then an MVP followed by a menu of design, development, and evaluation challenges—moves students from “I have no idea what to build” to shipping and evaluating real software.

Invented a way to grade open-ended projects with AI. When every student builds something different, conventional autograding is impossible and manual grading is infeasible at scale. I built a transparent **time-on-task grading** system in which a large language model reads each student's coding-agent session logs to estimate genuine active development effort (20 hours of effort earns full credit), making creative, individualized projects gradable at 2,000-student scale.

Kept classical programming rigorous. Agentic development complements rather than replaces classical programming: proctored quizzes remain 70% of the grade, with **36 proctored programming challenges across 14 weekly quizzes**—far more supervised coding than most introductory courses require—so students learn both to write code themselves and to direct an agent.

Created a course about AI, built with AI, and taught with AI. [Using and Understanding AI](#) (CS 199 UAI) is a new course for non-technical students, helping them build working mental models of how AI systems work and the judgment to use them well. Its entire platform—website, assessment system, in-class activity framework, course assistant, API server, database, and deployment infrastructure—was built **end-to-end through conversational programming with Claude Code**. Piloted in 2026, it is becoming a permanent course, **CS 108: Exploring Generative AI (2027)**.

Built a novel two-agent conversational AI assessment. In place of multiple-choice quizzes, students complete **chat-based oral exams conducted by AI**. A two-agent architecture separates a rubric-tracking *evaluator* from a conversational *interviewer*; the system distinguishes understanding demonstrated independently from understanding only surfaced after prompting, and can evaluate open-ended positions, not just recall. The same system runs unchanged in a proctored computer-based testing facility, and is validated by an adversarial test harness of synthetic student personas. One assessment is open for anyone to try.

Leading a new Applied Computing degree. I am leading the design of a proposed four-year undergraduate degree spanning four schools at Illinois—the Siebel School of Computing & Data Science, the School of Art & Design, the Gies College of Business, and the iSchool—organized around studio-based design, domain integration, and AI-assisted building.

Recognized for this leadership beyond my institution. I was selected as a **Replit Faculty Fellow**—an invited fellowship from [Replit](#), a globally used AI software-development platform—and as a **Faculty AI Fellow** in the University of Illinois [Summer AI Studio](#), where I provide disciplinary guidance and peer mentorship to faculty across the university integrating AI into their teaching.

Teaching at Scale and Open Infrastructure

Transformed CS1 at one of the world’s leading computer science programs. Since 2017 I have redesigned and led [CS 124](#), the introductory programming course at the University of Illinois (consistently ranked among the top five CS departments worldwide). The course now enrolls **over 2,000 students per year** (≈80% non-majors). Since I took it over, enrollment has doubled, drop rates have fallen, success rates have risen, performance in later courses has improved, and a gender performance gap has disappeared.

Built and openly released the infrastructure behind it, with international reach. I personally engineer the systems that make interactive computing education work at scale, and publish them as free and open resources:

- **[learnes.online](#)** — a freely available version of my CS1 course: 61 daily lessons in Java and Kotlin, 902 runnable code playgrounds, **2,540 interactive walkthroughs contributed by 264 instructors**, and 700+ autograded programming problems. In its first year it supported **thousands of independent learners worldwide**, who generated **hundreds of thousands of homework submissions**. It is adopted beyond Illinois—used by Ellen Spertus at Northeastern University for her Kotlin course, and as the interactive textbook for a high-school outreach program through Chicago’s Discovery Partners Institute.
- **[Jeed](#)** — an open-source toolkit that executes untrusted Java/Kotlin code in a secure sandbox **1000× faster** than container-based alternatives, in production since 2019 and used by other educators.
- **[Questioner](#)** — a novel solution-driven authoring system that generates and validates autograders, producing accurate programming exercises an **order of magnitude faster** than prior approaches; being released for broader use at [beyondgrader.com](#).
- **[ops-class.org](#)** and **[internet-class.org](#)** — open course frameworks (operating systems; how the internet works), including the `test161` autograder and hundreds of hours of openly available lecture video.

Recognized leadership in the field. Publications at **SIGCSE** (the premier international CS-education venue) in 2024 and 2025; an **invited keynote** at the Western Canadian Conference on Computing Education (2022); co-founder of the Illinois Computer Science Summer Teaching Workshop (now in its fifth year); and an active member since 2021 of an international community of practice of 100+ CS educators from dozens of institutions worldwide.

Research Record (*evidence of international standing*)

From 2011–2017 I was a tenure-track Assistant Professor at the University at Buffalo, leading the **blue Systems Group** in mobile-systems research; my Ph.D. work at Harvard (2004–2010) helped establish the field of wireless sensor networks.

- **More than 8,000 citations; h-index 22** (Google Scholar), spanning mobile systems and sensor networks.
- Publications in the **top international venues** in the field: OSDI, SenSys, MobiSys, UbiComp, INFOCOM, IPSN, HotOS, and HotMobile.
- **Best Paper Award**, Runtime Verification (RV'16); **Best Poster Award**, HotMobile'14.
- Led **PhoneLab**, an NSF-funded public smartphone testbed (≈ 150 participants) used by researchers internationally.
- Foundational sensor-network work, including deploying wireless sensor networks on **active volcanoes** (OSDI'06) and the widely used **MoteLab** testbed (IPSN'05).
- **International and industry collaborations**: CSIRO (Australia), TU Darmstadt (Germany), Microsoft Research, and AT&T Labs Research; co-editor of a Springer volume on wireless sensor networks (2010).

Recognition

- **2026** — Replit Faculty Fellow (invited fellowship from Replit, a globally used AI software-development platform)
- **2026** — Faculty AI Fellow, Summer AI Studio, Center for Innovation in Teaching & Learning, University of Illinois
- **2026** — Rose Award for Teaching Excellence, The Grainger College of Engineering, University of Illinois
- **2026** — Scott H. Fisher Computer Science Teaching Award (the Siebel School's top undergraduate-teaching honor)
- **2017** — National Science Foundation **CAREER Award**
- **2015** — SEAS Early Career Teacher of the Year, University at Buffalo
- **2009** — Siebel Scholar

Selected Publications

Computer science education

- Y. Zhou, L. Paquette, **G. Challen**. *Investigating the Presence and Development of Student Instructor Preferences in a Large-Scale CS1 Course*. SIGCSE 2025.
- **G. Challen**, B. Nordick. *Accelerating Accurate Assignment Authoring Using Solution-Generated Autograders*. SIGCSE 2025.
- H. Chen, A. Li, **G. Challen**, K. Cunningham. *Implementation of Split Deadlines in a Large CS1 Course*. SIGCSE 2024.
- L. Butler, **G. Challen**, T. Xie. *Data-Driven Investigation into Variants of Code Writing Questions*. IEEE CSEE&T 2020.
- J. Osei-Owusu, A. Astorga, L. Butler, T. Xie, **G. Challen**. *Grading-Based Test Suite Augmentation*. ASE 2019.

Mobile systems and sensor networks

- J. Shi, S. Lahiri, R. Chandra, **G. Challen**. *Wireless Protocol Validation Under Uncertainty*. Formal Methods in System Design, 2017 (*journal version of RV'16 Best Paper*).
- A. Nandugudi, T. Ki, C. Nuessle, **G. Challen**. *PocketParker: Pocketsourcing Parking Lot Availability*. UbiComp 2014.
- K. Lorincz, B. Chen, **G. Challen**, et al. *Mercury: A Wearable Sensor Network Platform for High-Fidelity Motion Analysis*. SenSys 2009.
- **G. Werner-Allen**, K. Lorincz, J. Johnson, J. Lees, M. Welsh. *Fidelity and Yield in a Volcano Monitoring Sensor Network*. OSDI 2006.
- **G. Werner-Allen**, P. Swieskowski, M. Welsh. *MoteLab: A Wireless Sensor Network Testbed*. IPSN 2005.

A full publication list (40+ peer-reviewed papers) is available on Google Scholar and at <https://bluegroup.systems/papers>.

Research Funding

Almost **\$3M in competitive external funding** across my career, primarily from the National Science Foundation and Google, most as PI. Selected awards:

- **NSF CAREER:** Harnessing Implementation Flexibility to Enable Runtime Adaptation, \$525,582 (2017), sole PI.
- **NSF:** PhoneLab — a Programmable Participatory Smartphone Testbed, \$1,322,510 (PI).
- **NSF:** Jouler — Cross-Device Application Energy Management for Smartphones, \$499,185 (sole PI).
- **NSF:** Architecture and System Support for Power-Agile Computing, \$561,766 (co-PI).
- Additional NSF and Google awards for smartphone-testbed and pocket-scale data-management research, and two Illinois Strategic Instructional Innovations Program (SIIP) awards for CS1 redesign.

Professional Recognition and Service

- **Editor**, *GetMobile* (ACM SIGMOBILE magazine), 2015–.
- **Program committees / reviewing** for top international venues: MobiCom, MobiSys, SenSys, IPSN, HotMobile, and others; first University at Buffalo faculty member invited to serve on several of these.
- **Invited talks** include *The Educational Engineer*, University of Sydney (2026), surveying my CS-education and AI-era work; and the WCCCE keynote (2022).
- **Invited participant:** White House SmartAmerica Challenge (representing PhoneLab); Google Faculty Summits (2011, 2015); Visiting Researcher, IT University of Copenhagen (2010).
- Founded and advised the University at Buffalo chapter of the **Scientista Foundation** (women in STEM) and led a delegation to the Grace Hopper Celebration.

Teaching and Mentorship

- Over **21 terms at Illinois I have taught 16,107 students**, the large majority in the CS1 course (CS 124, formerly CS 125) I rebuilt and lead; enrollment has roughly doubled to **over 2,000 students per year**.
- Designing and launching new courses on AI for non-technical students—*Using and Understanding AI* (CS 199 UAI / CS 108) and *Conversational Programming* (CS 199 ASK)—alongside the proposed Applied Computing degree.
- At the University at Buffalo I graduated one Ph.D. student and **supervised 47 graduate and undergraduate researchers** across Ph.D., Masters, and undergraduate levels; at Illinois I recruit, train, and mentor a large course staff each term.

Appointments and Education

- **2027–** Associate Professor (Level D), University of Sydney (*forthcoming, from May 2027*)
- **2024–** Teaching Professor, University of Illinois Urbana-Champaign
- **2017–2024** Associate Teaching Professor, University of Illinois Urbana-Champaign
- **2011–2017** Assistant Professor, University at Buffalo
- **2010–2011** Postdoctoral Associate, MIT (advised by Hari Balakrishnan)
- **2010** Ph.D., Computer Science, Harvard University (advised by Matt Welsh)
- **2003** A.B., Physics, Harvard University